

Projeto Estádio do Dragão – Fan App

M2: prototype

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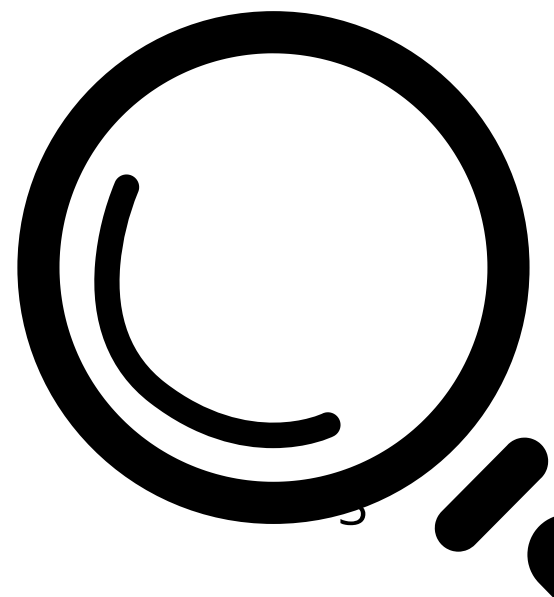
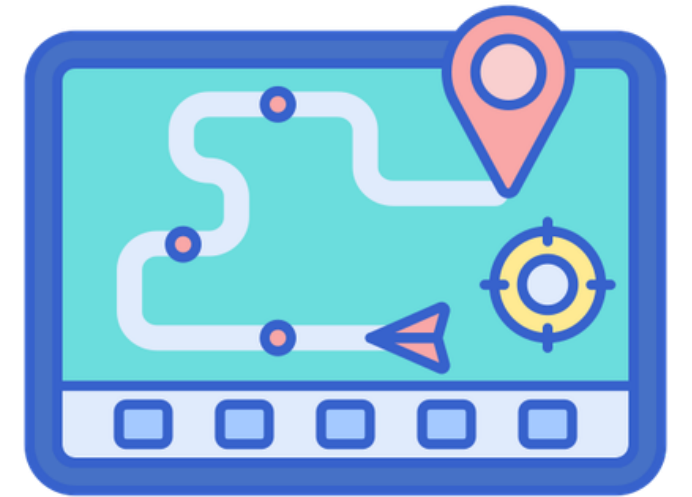
Abdelgader Mahmoud

Overview

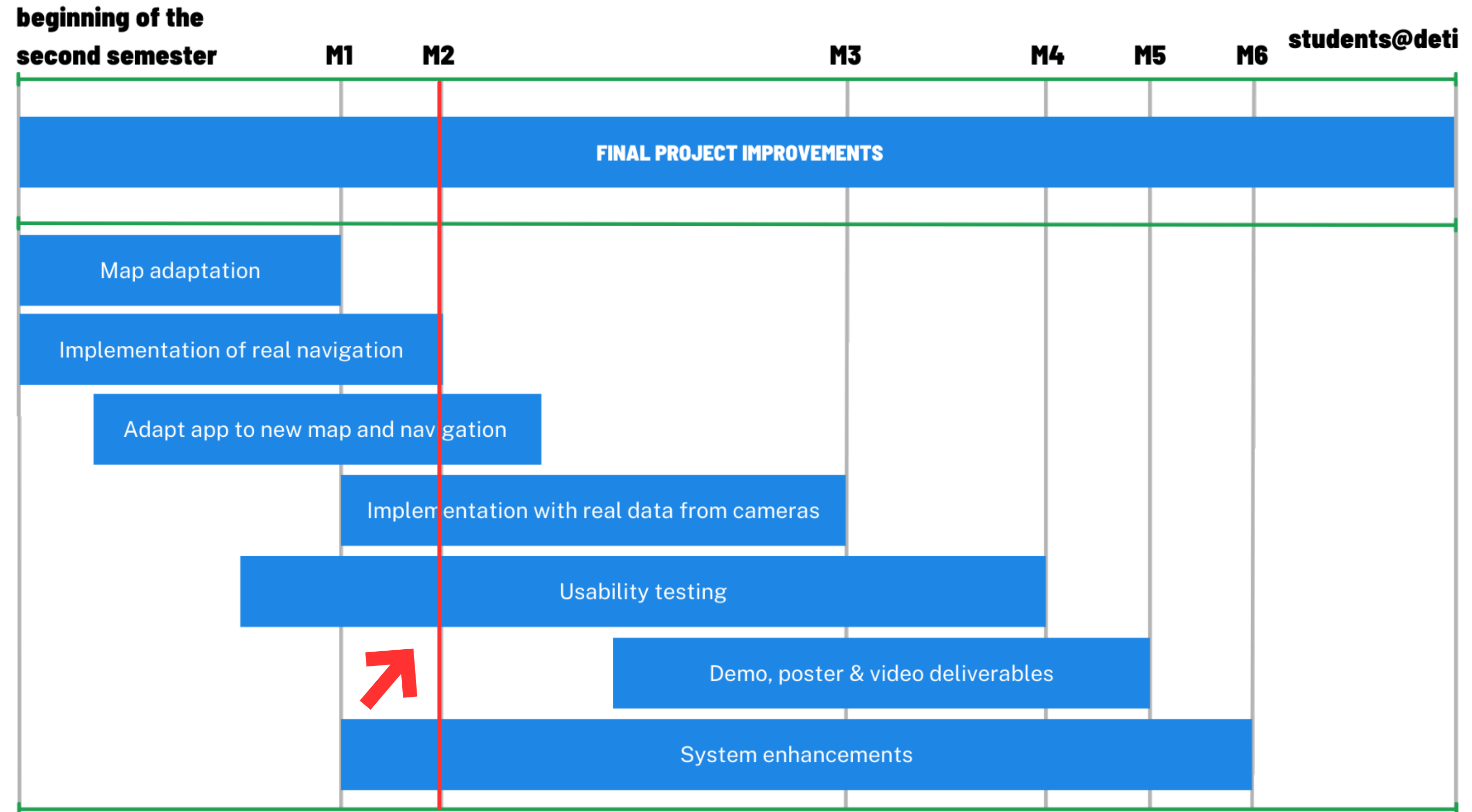
- 01** Context and Goals
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- 03** Personas (dashboard)
- 04** Architecture (dashboard)
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- 07** Personas (Fanapp)
- 08** Architecture (Fanapp)
- 09** Demo (Fanapp)

Context and Goals

- Improve campus navigation for students, staff, and visitors
- Provide real-time outdoor navigation
- Ensure safety with accessible and emergency evacuation routes
- Provide real-time congestion's maps and queues of UA



Context and Goals



Graph Editing Dashboard

Needs

- Node Graph easy customization
- Service hosts being able to make quick adaptations due to changes in the campus ground or for events



Persona

Graph Editing Dashboard



Fanapp service hoster

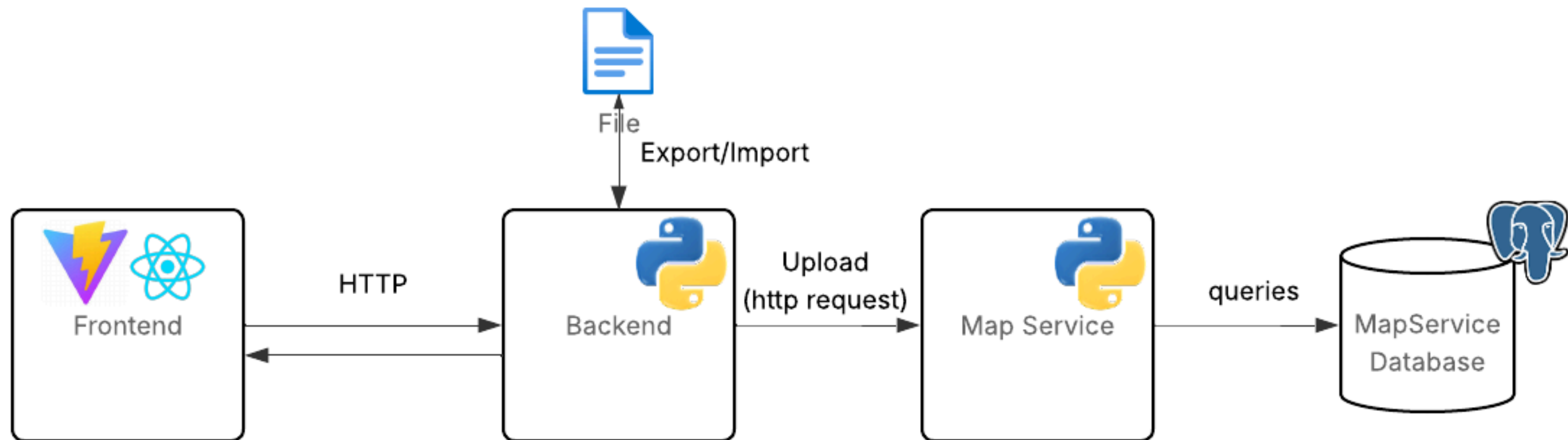
Role: Manages Fanapp services and ensures the application maps are updated and working correctly.

Needs: Reliable map data, and quick access to fix map-related issues.

Objective: Ensure users can view locations and events on the map without errors, failures or discrepancies.

Architecture

Graph Editing Dashboard



Demonstration

Graph Editing Dashboard

- Selection
- Add nodes
- Edit nodes
- Delete nodes
- Add edges
- Delete edges
- Bulk delete
- Edge filter
- Import
- Export
- Upload

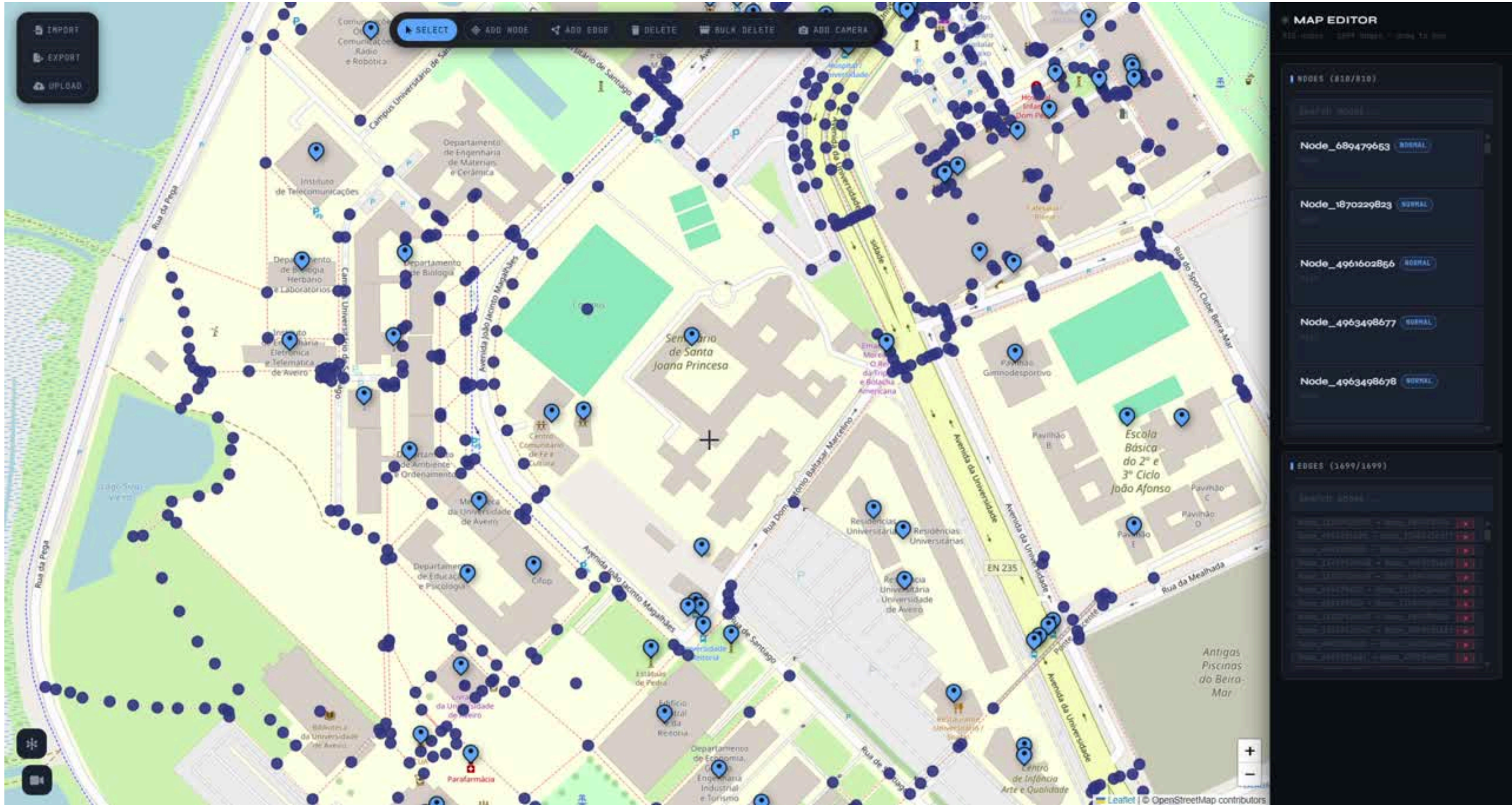
Selection and node edition

Graph Editing Dashboard Demo

- As a Fanapp service hoster,
- I want to be able to select and edit the map nodes
- So that, the users will have up to date and consistent with in real life POIs and intersections

Selection and node edition

Graph Editing Dashboard Demo



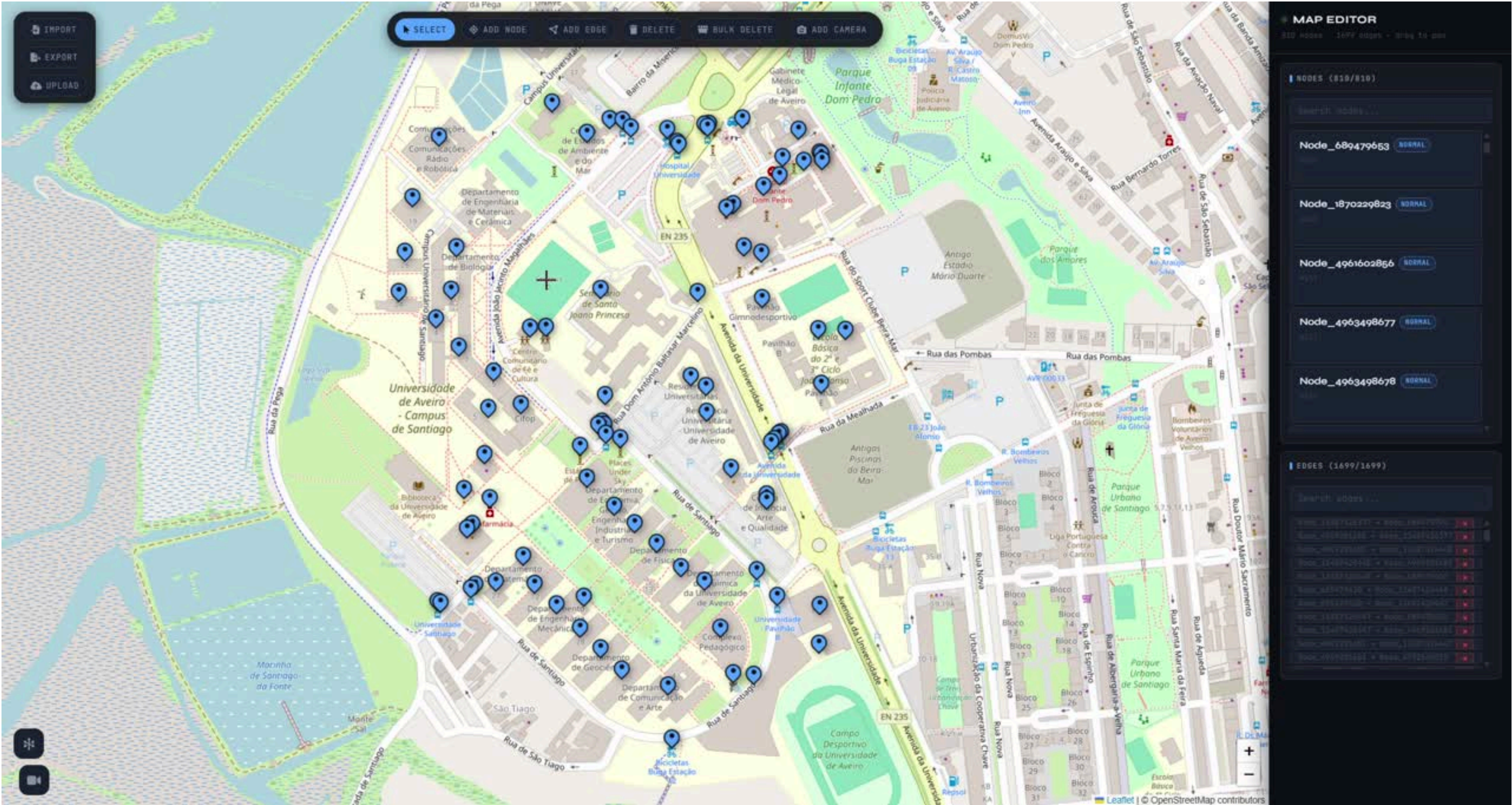
Add Nodes

Graph Editing Dashboard Demo

- As a Fanapp service hoster,
- I want to be able to add nodes to the map
- So that, the users will have up to date and consistent with in real life map POIs/ intersections or event-specific map locations

Add Nodes

Graph Editing Dashboard Demo



Show edges and Delete

Graph Editing Dashboard Demo

- As a Fanapp service hoster,
- I want to be able to see all the edges (paths) and delete the unnecessary one, as well as the unnecessary nodes
- So that, the users will have up to date paths and no useless POIs/intersections

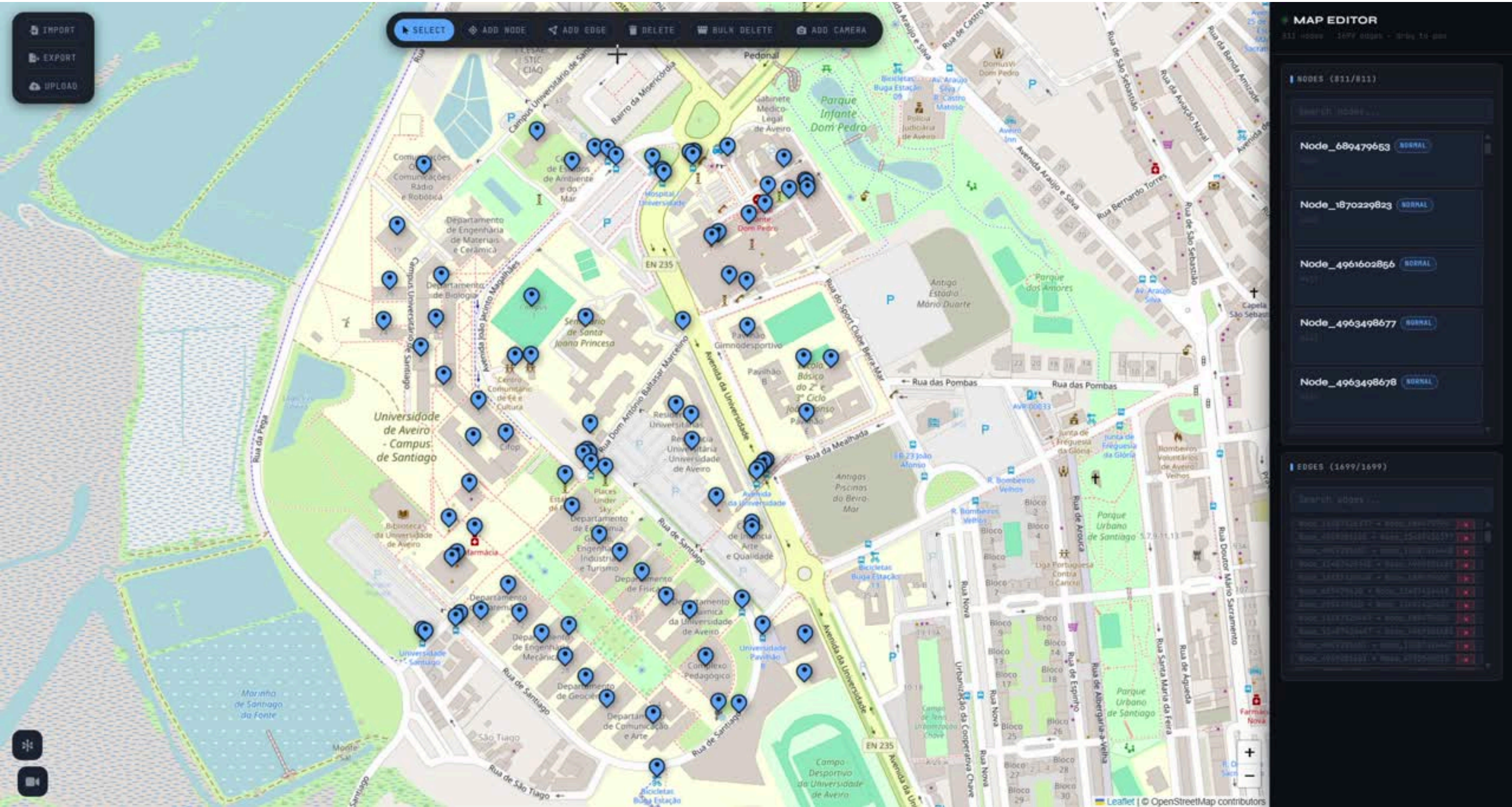
Add Edges

Graph Editing Dashboard Demo

- As a Fanapp service hoster,
- I want to be able to add edges to the map
- So that, the users will have up to date or event-specific paths

Add Edges

Graph Editing Dashboard Demo



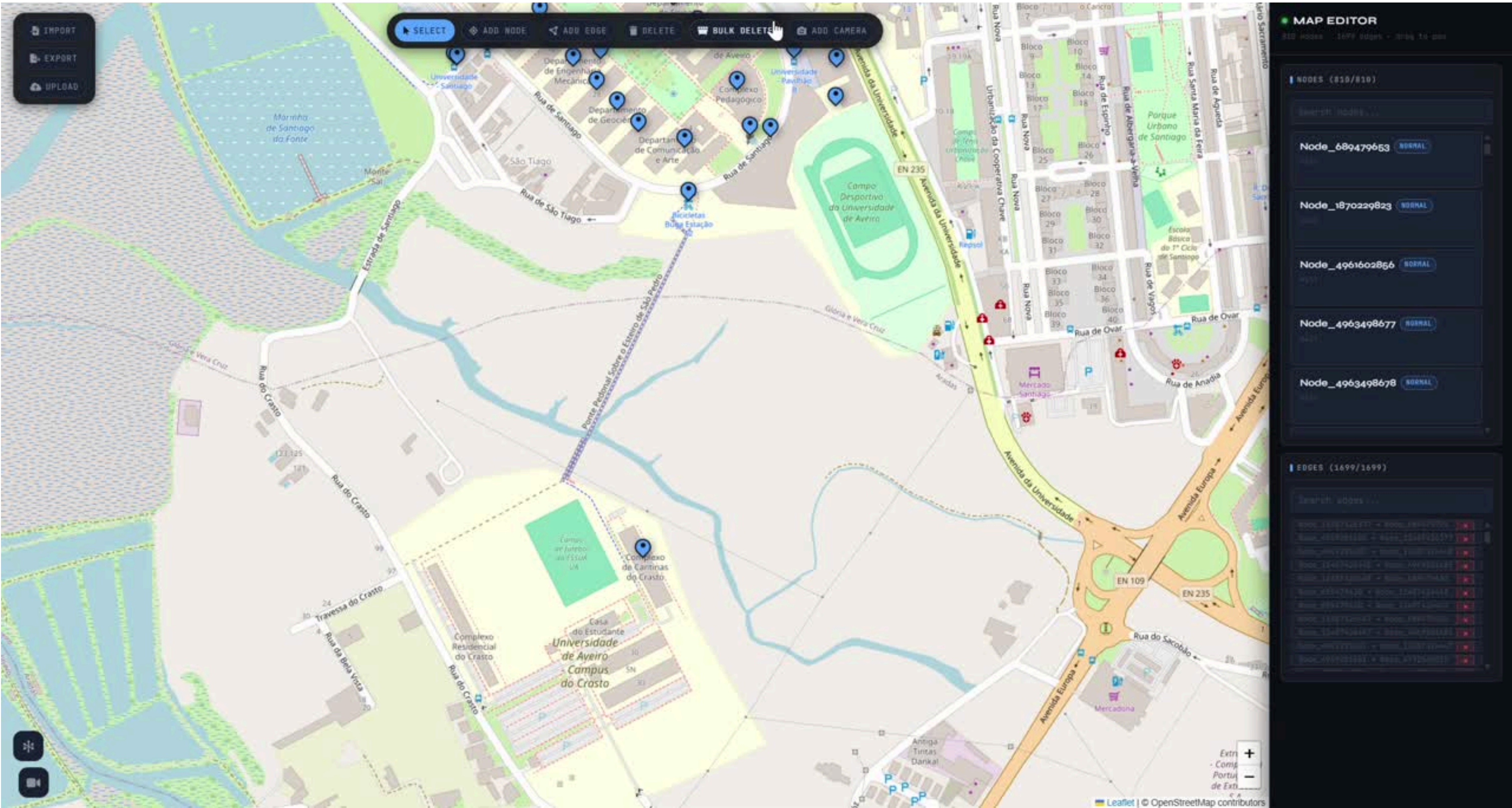
Bulk Delete

Graph Editing Dashboard Demo

- As a Fanapp service hoster,
- I want to be able to delete grand quantities of nodes and edges
- So that, i don't waste time deleting them one by one when big changes in the map occur

Bulk Delete

Graph Editing Dashboard Demo



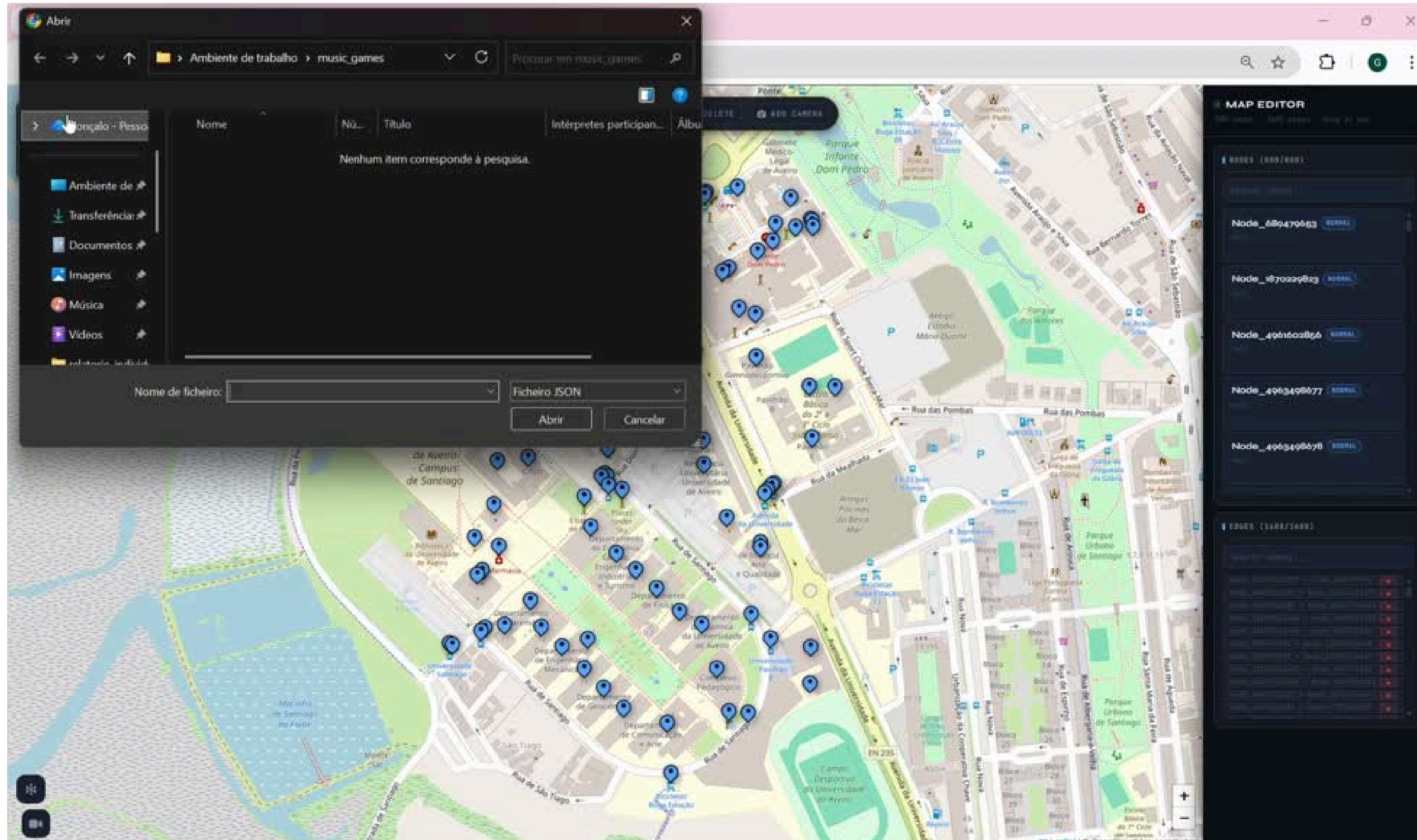
Import, Export and Upload

Graph Editing Dashboard Demo

- As a Fanapp service hoster,
- I want to be able to import, export to a file and upload into the map-service database the editions to the map
- So that the map can be easily managed and worked my multiple hosts and so users have an up-to-date map

Import, Export and Upload

Graph Editing Dashboard Demo



Fan App

Needs

- In-campus navigation and POI finding
- Congestion-aware routing, services (queues) and heat-maps
- POI filtering
- Emergency evactions
- Accessibility features



Personas

Fanapp

Campus regulars



Alberto

Campus regular



Argelino

Accessibility-
Focused regular



Josue

Safety-Aware
regular

Alberto: Campus regular



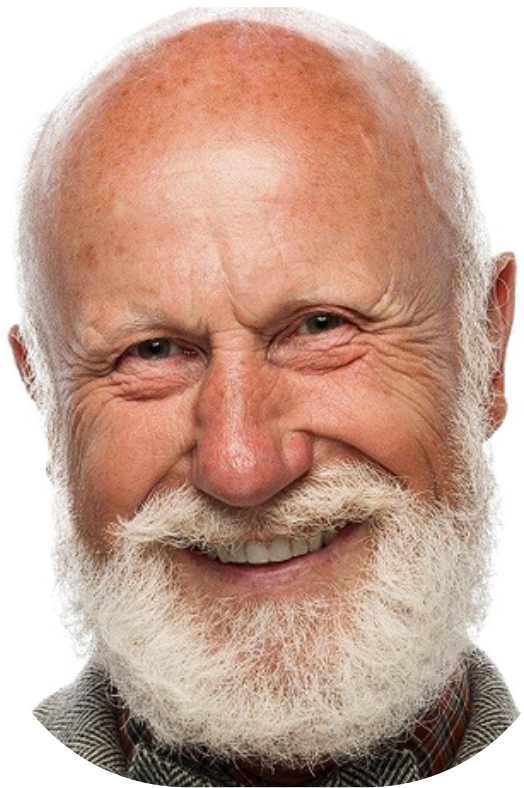
Role: Regular campus user who often arrives late and needs to move quickly between locations.

Objective: Find recommended paths that avoid long queues and reach his department faster.

Needs: Real-time route suggestions, queue information, and clear campus navigation.

Argelino

Accessibility-Focused regular



Role: International campus user who does not speak Portuguese, who also has issues with mobility and over-stimulation

Objective: Use the app in his language to understand instructions and navigate the campus through accessible paths and while avoiding crowds

Needs: Multi-language support accessible routing mode and congestion-aware navigation

Josue

Safety-Aware regular



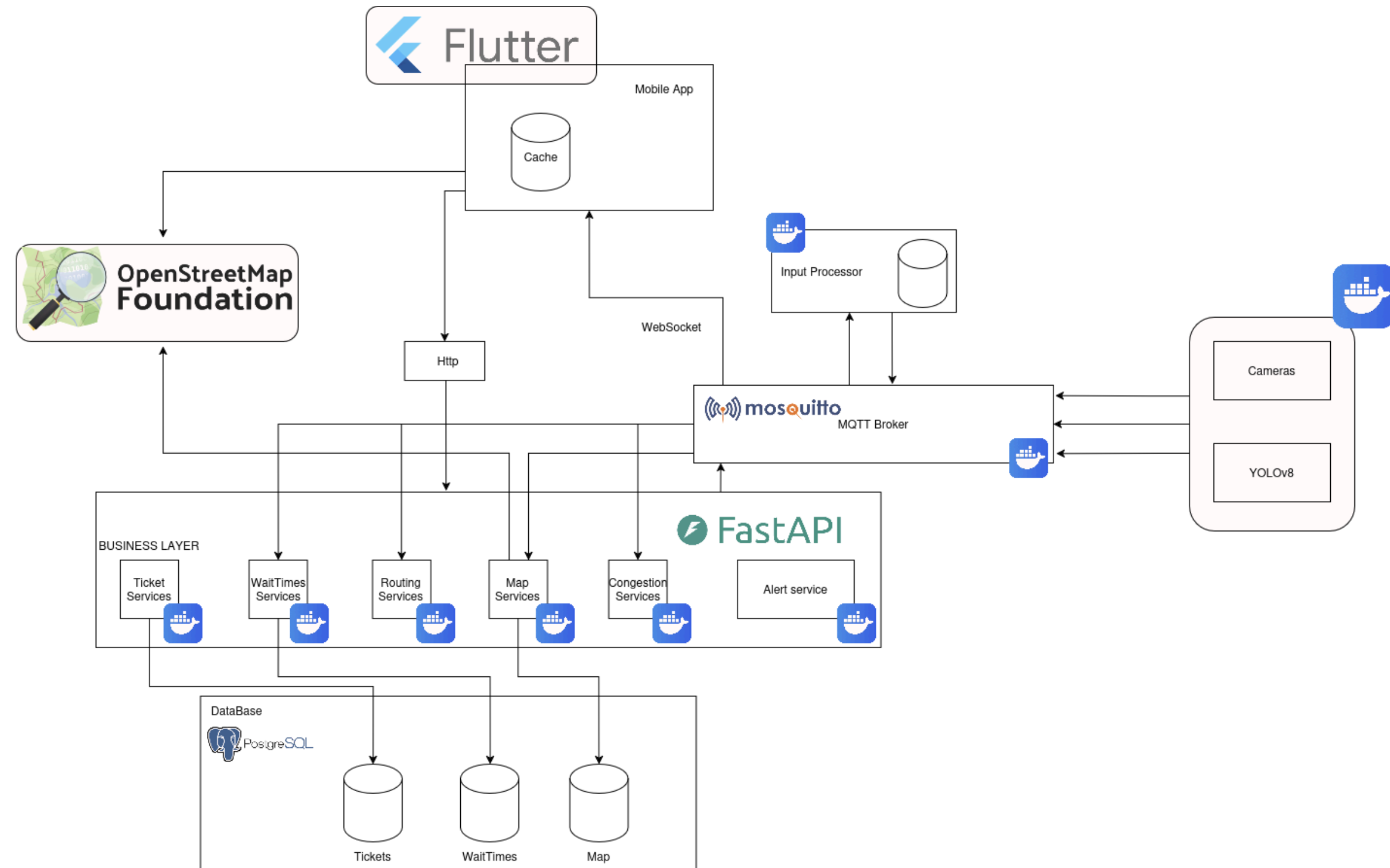
Role: Campus user who prioritizes safety while moving around the campus.

Objective: Receive fast notifications in case of emergencies.

Needs: Real-time alerts, clear safety instructions, and reliable notifications.

Architecture & Tech

Fanapp



Changes & new features

Fanapp

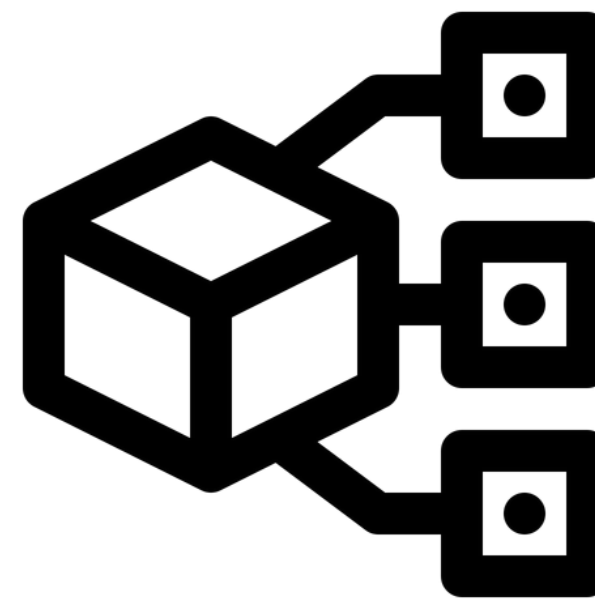
- Implementation of the waitimes-queue, that is using a video in a loop, detects the people and estimates a queue time.
- Minor changes to UI to match the usability feedback
- Routing and re-routing improvements to make route-finding faster



Prototype Demo

Fanapp

- Simple navigation
- Navigation with re-routing
- Filtering
- Wait-time
- Accessibility Switch
- Search-bar
- Heat-map
- Favourites
- Alerts



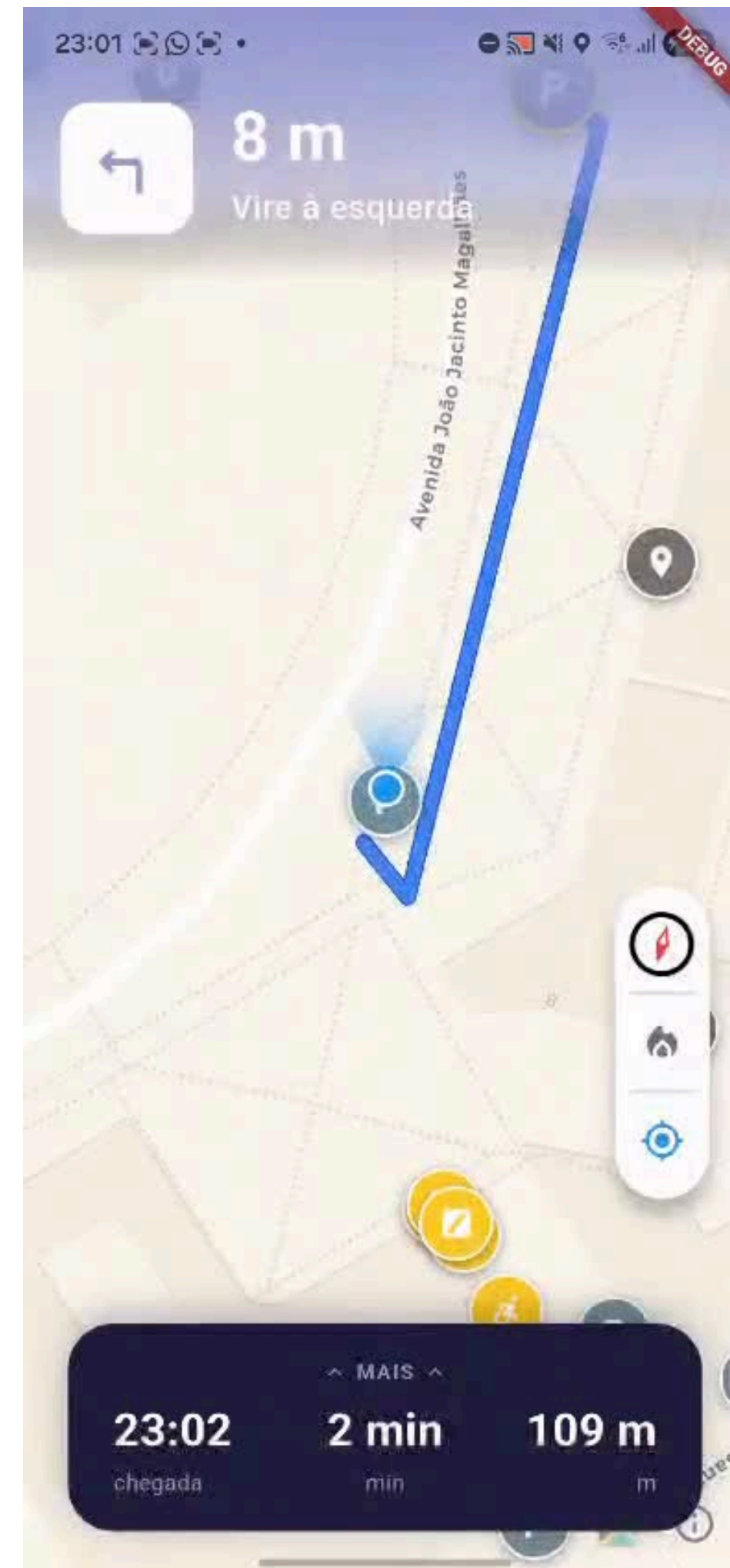
System Demonstration

Simple navigation

- As a Fanapp campus frequenter
- I want to be able to select a POI and get an optimal route in order to make my way over there
- So that i can reach a destination as fast as possible

System Demonstration

Simple navigation



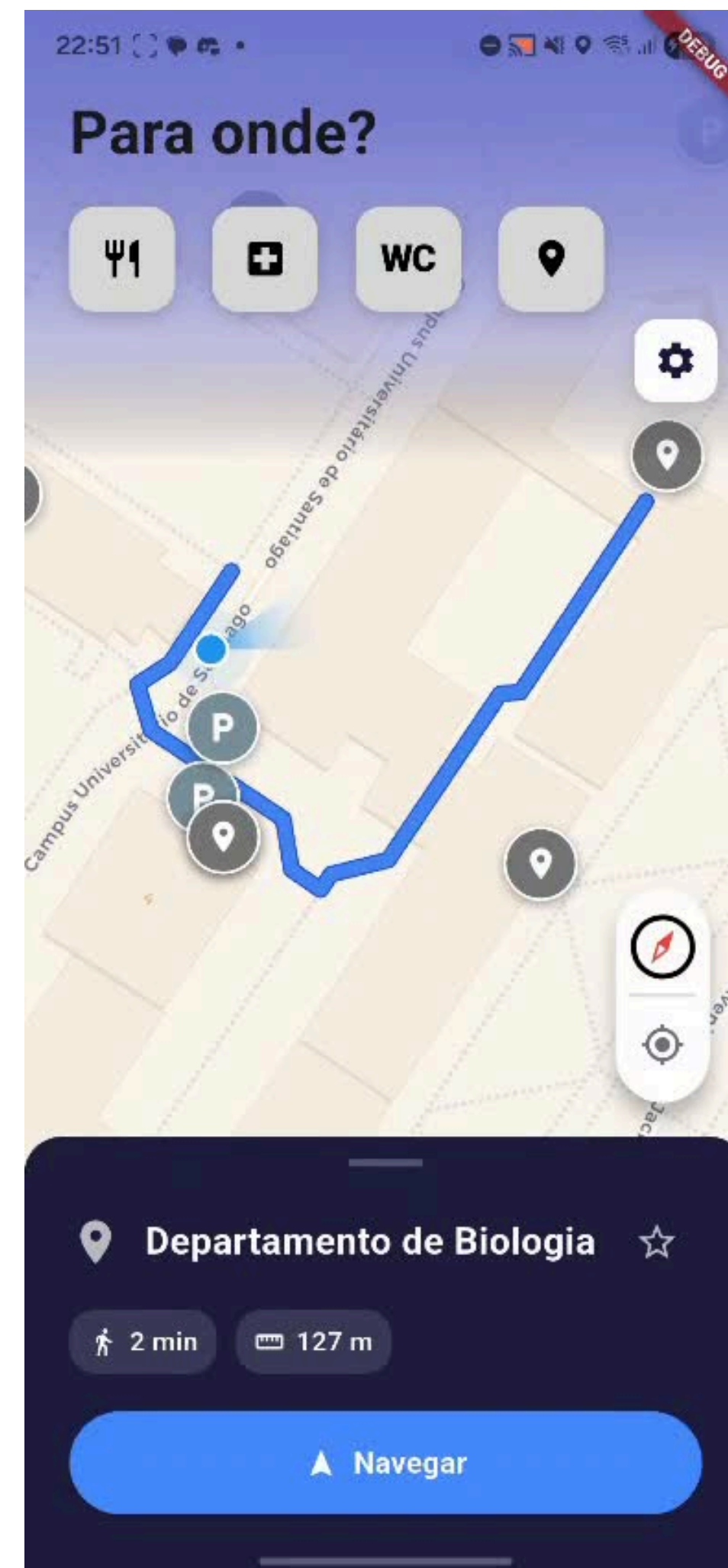
System Demonstration

Navigation with Re-
routing

- As a Fanapp campus frequenter
- I want my routes in navigation mode to automatically update based on either congestion or me leaving the original route
- So that if I unintentionally or intentionally take a detour, i can still go to the right destination

System Demonstration

Navigation with Re-
routing



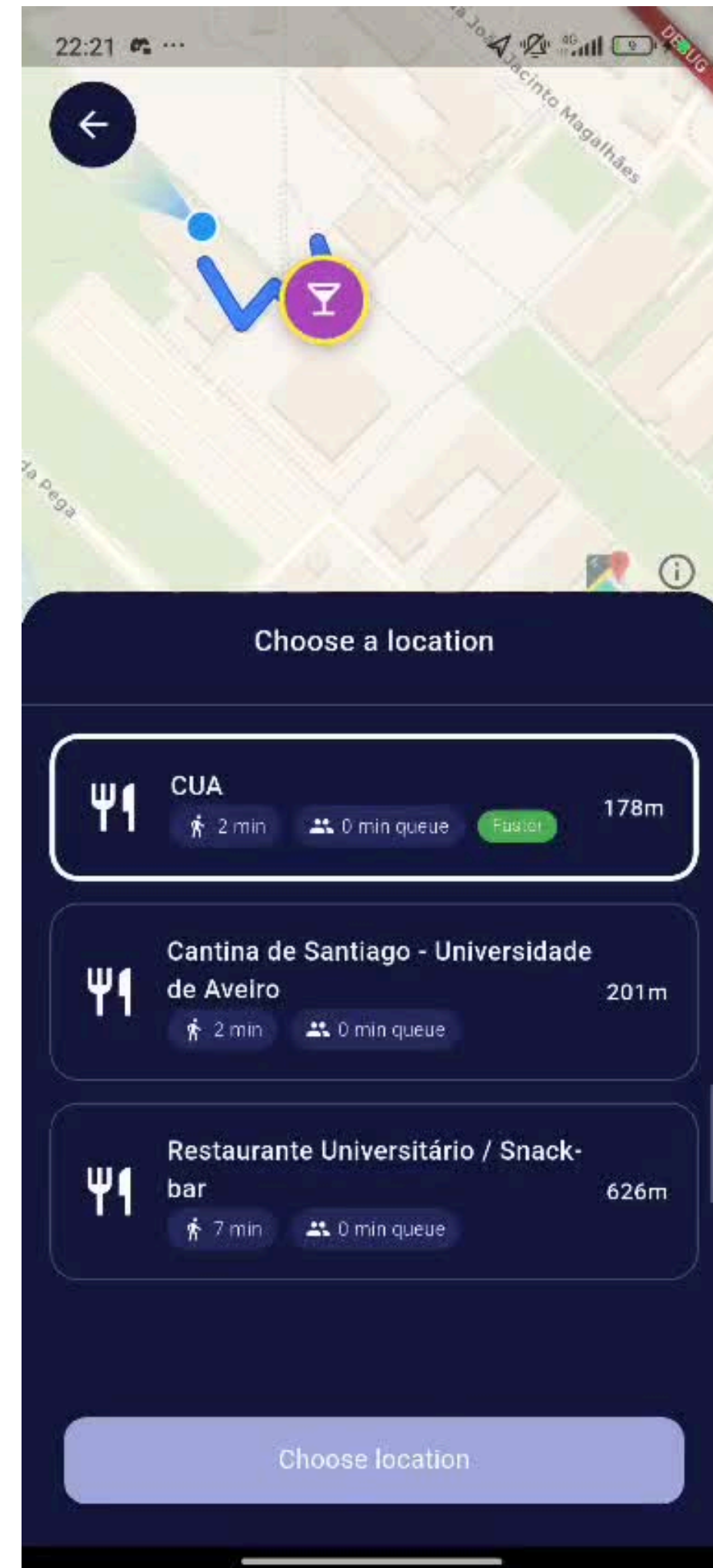
System Demonstration

Filtering

- As a Fanapp campus frequenter
- I want to be able to view a menu of options for a specific kind of service/POIs
- So that i can find a service that offers me what I want quickly

System Demonstration

Filtering



System Demonstration

Wait-times

- As a Fanapp campus frequenter
- I want to receive estimated queue times for services when i select a service POI
- So that I can make informed decisions when I should visit certain services

System Demonstration

Wait-times



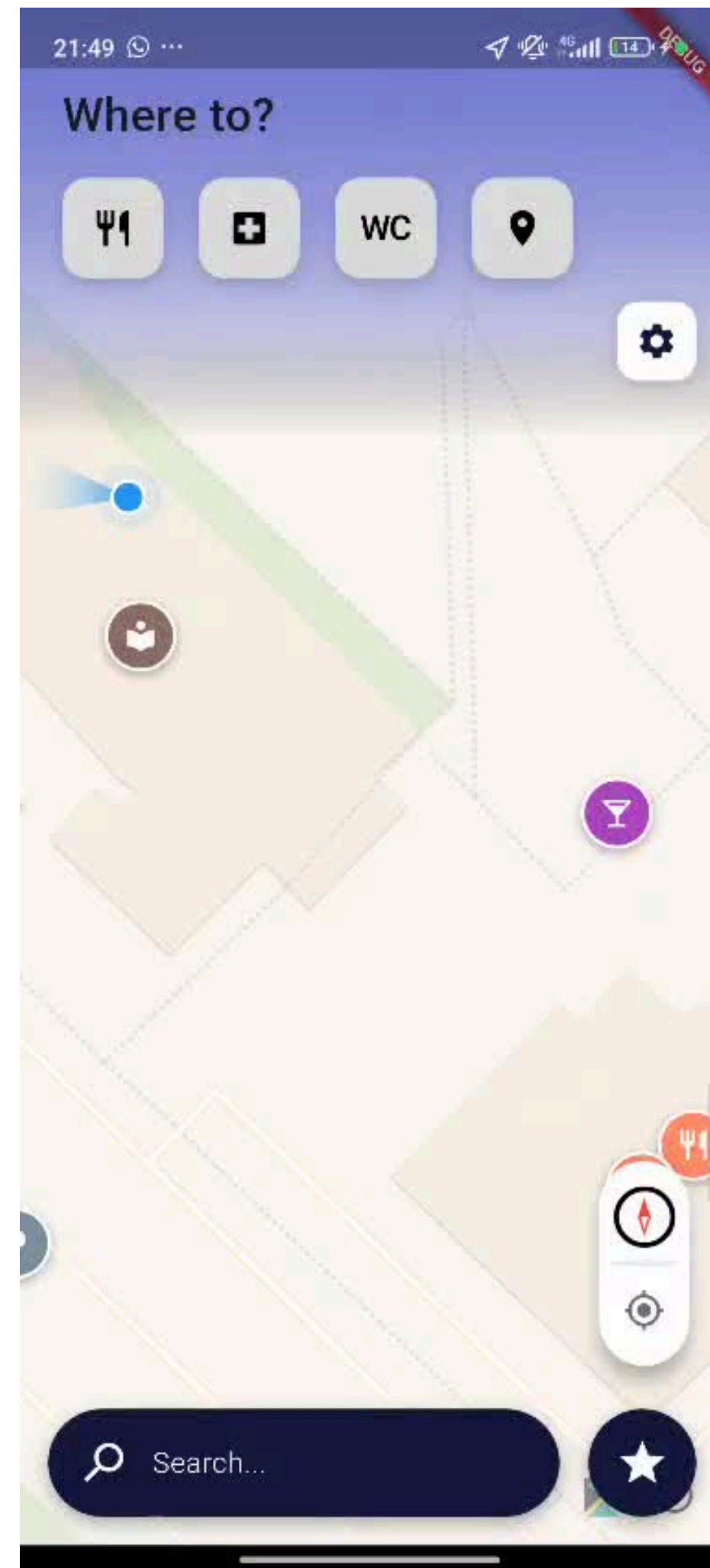
System Demonstration

Accessibility switch

- As an accessibility-focused campus regular
- I want to be able to click a switch on the Fanapp that makes routing only take accessible routes
- So that I only get paths that I'm able to travel through

System Demonstration

Accessibility switch



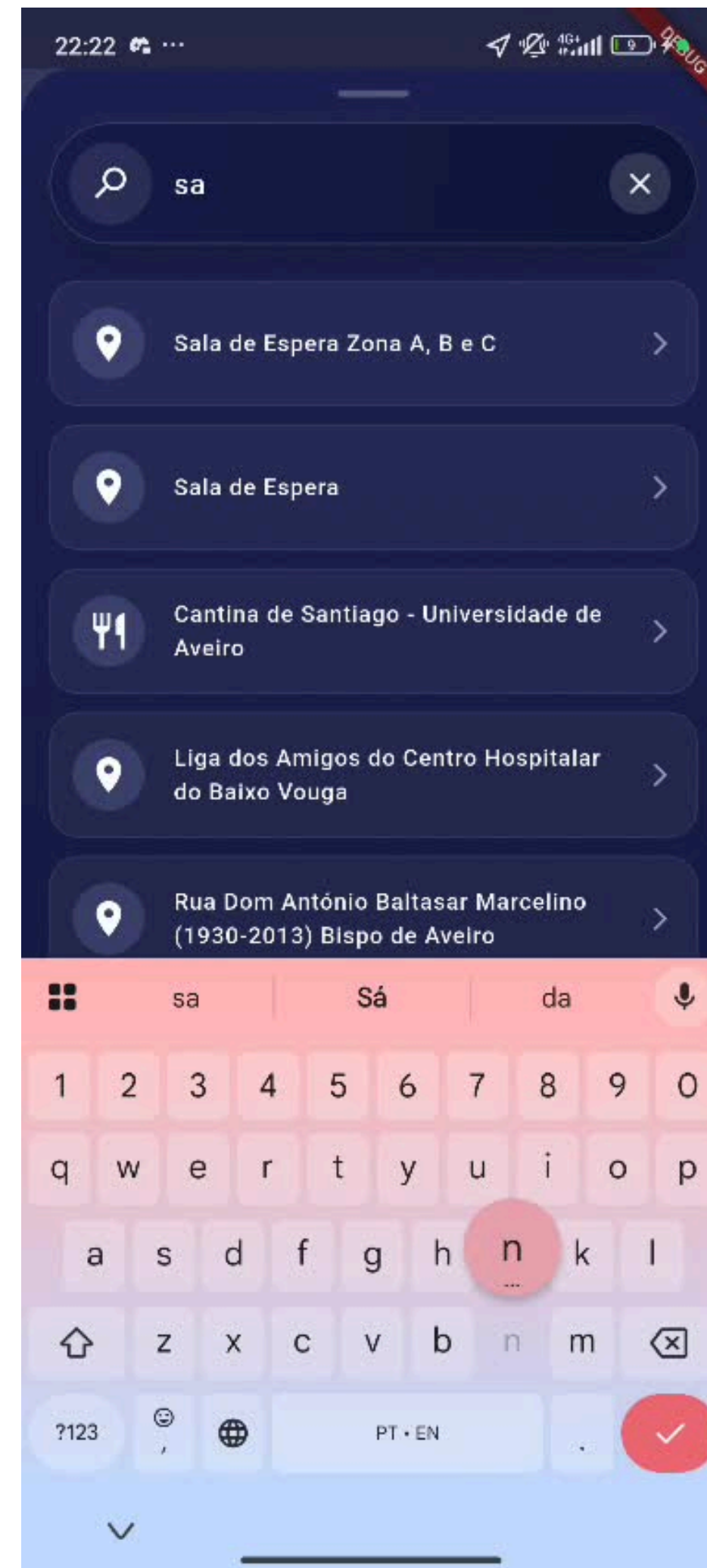
System Demonstration

Search bar

- As a Fanapp campus frequenter
- I want to be able to search for a specific POI
- So that i can find it faster and retrieve information about it or enter navigation mode to get to it

System Demonstration

Search bar



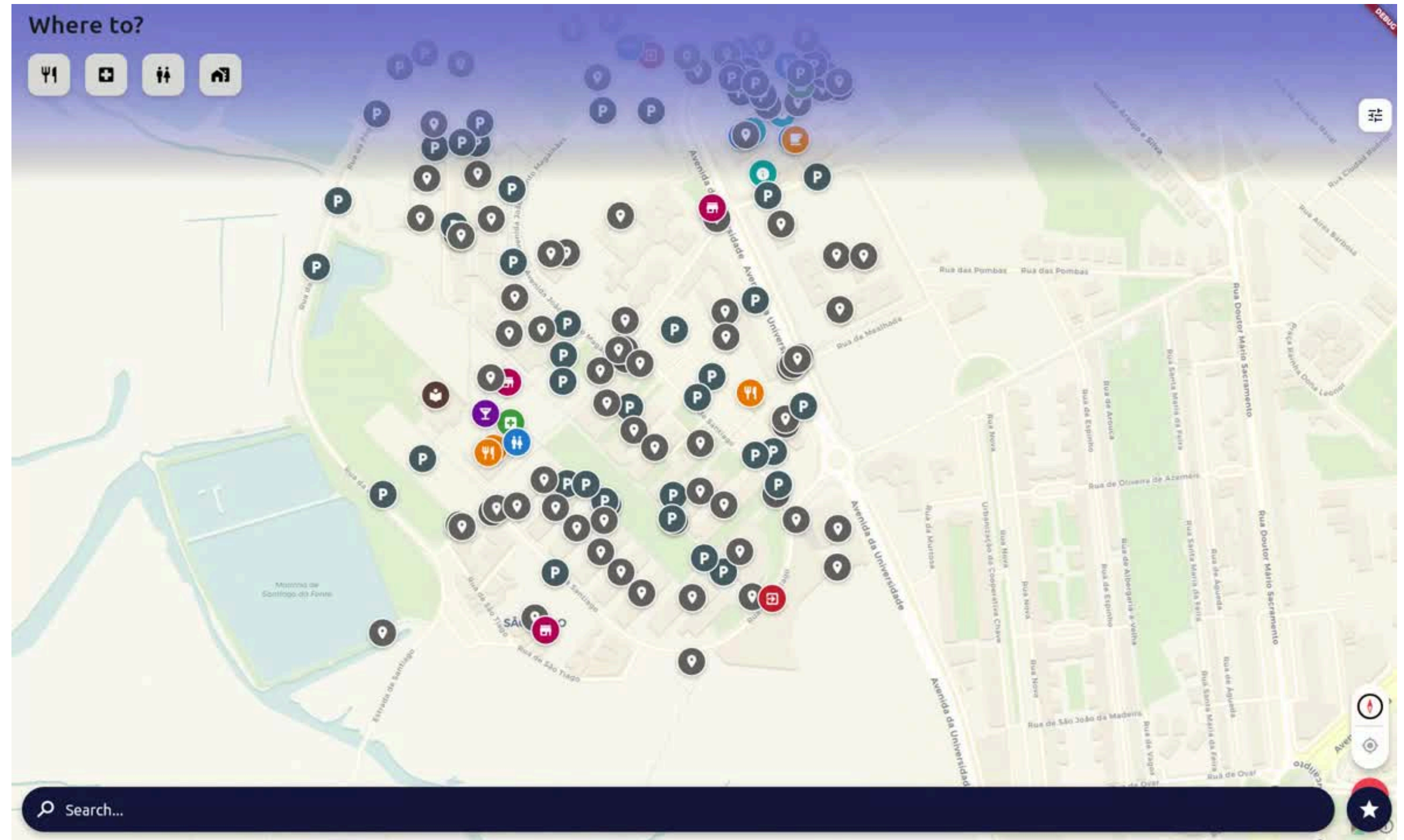
System Demonstration

Heat-map

- As an accessibility-focused campus regular
- I want to be able to see live congestion data
- So that i can avoid locations with a lot of noise/
stimulation

System Demonstration

Heat-map



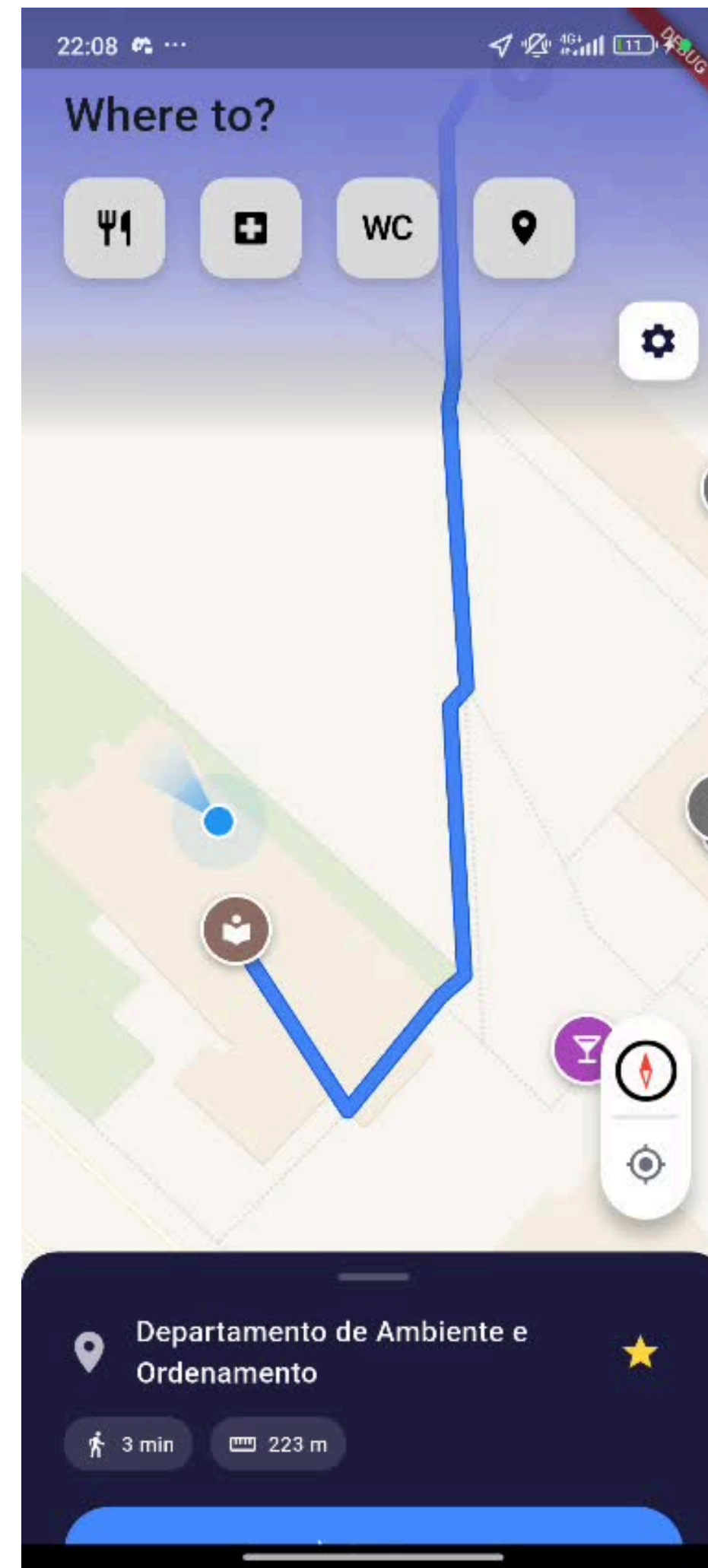
System Demonstration

Favourites

- As an campus frequenter
- I want to be able favourite and unfavourite POIs
- So that i can consistently quickly select my preferred locations

System Demonstration

Favourites



System Demonstration

Alert

- As a security-focused campus regular
- I want to receive an alert in-app
- So that i can quickly be advised of danger and know the quickest way out in this panic situation

System Demonstration

Alert



Questions?



Thank You!

Project Microsite:

